



MTN Liquidation API Doc V 1.1

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DEVELOPER API GUIDE DOCUMENT

PROJECT: Equity IMT Outbound On Premise

API Document

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1.0 NAME CHECK SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/namecheck
Headers	Content-Type:application/xml

1.0.1 Name Check Request Fields And Description

NO	FIELD	DESCRIPTION	MANDATORY	TYPE
1	mobileNumber	Destination mobile. Allowed Mobile Number Length Is 12 Characters Excluding Plus (+) Suffix. Example - 256779000000	Yes	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Yes	String
3	msgId	Reference number	Yes	String
4	telcom	Name of the destination Telco Example: MTN-UG	Yes	String
5	channel	Channel code Example: MFS	Yes	String
6	signature	Digital signature. See Digital signature section for more information.	Yes	String
7	nationalID	Id of the receiver. If not, supply 000000	Yes	String

1.0.2 Name Check Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	msgId	Reference number	String
2	channel	Channel code	String
3	mobileNumber	Destination mobile.	String
4	responseCode	Response code	String
5	customerNames	Unique identifier for the transaction	String

1.0.3 Name Check Status Possible Values

RESPONSE CODE	DESCRIPTION
00	Name check request successful
01	Name check request failed

1.0.4 Name Check Sample Request

```
<?xml version="1.0" encoding="UTF-8"?>
<mobileNumberInquireReq>
  <mobileNumber>256779999703</mobileNumber>
  <timestamp>26-Aug-2022 12:55:32</timestamp>
  <msgId>415976</msgId>
  <telcom>MTN-UG</telcom>
  <channel>MFS</channel>
  <signature>QmxK19mYaXgLx4+dWTxetKLdkFSD6SpF5gaPER41+Q==</signature>
  <nationalID>000000</nationalID>
</mobileNumberInquireReq>
```

1.0.5 Name Check Sample Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<MobileNumberInquireRes>
  <msgId>415976</msgId>
  <channel>MFS</channel>
  <mobileNumber>256779999703</mobileNumber>
  <responseCode>00</responseCode>
  <customerNames>Martha A</customerNames>
</MobileNumberInquireRes>
```

2.0 DEBIT/ DEPOSIT SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/deposit
Headers	Content-Type:application/xml

2.0.1 Debit Request Fields And Description

No	FIELD	DESCRIPTION	MANDATORY	TYPE
1	msgld	Reference number	Y	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Y	String
3	rrn	Retrieval reference number (Can be same as msgld)	Y	String
4	amount	Transaction amount rounded off to 2 decimal places. Example: 100.00	Y	String
5	currencyCode	Currency	Y	String
6	tranType	Type of Transaction. Example: WalletToAccount	Y	String
7	receiverMsisdn	Destination mobile. Allowed Mobile Number Length Is 12 Characters Excluding Plus (+) Suffix. Example - 256779000000	Y	String
8	receiverName	Name of the receiver. Usually from Name Check service	Y	String
9	senderName	Sender name	Y	String
10	senderMsisdn	Sender mobile. Allowed Mobile Number Length Is 12 Characters Excluding Plus (+) Suffix. Example - 256779000000	Y	String
11	callbackURL	Sending channel callback url. Used to send final transaction status to the calling channel.	Y	String
12	signature	Digital signature. See Digital signature section for more information.	Y	String

2.0.2 Debit Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	msgId	Reference number	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	String
3	rrn	Retrieval reference number	String
4	status	Transaction status	String
5	responseDesc	Transaction status description	String
7	transactionRef	Unique identifier for the transaction	String

2.0.3 Debit Status Possible Values

STATUS(status)	DESCRIPTION
ACK	Debit request was accepted by MTN awaiting the final status
FAILURE	Debit request failed

2.0.4 Sample Debit Request

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<EazzyDepositReq>
  <msgId>264375443725</msgId>
  <timestamp>2022-08-26T15:39:22.518</timestamp>
  <rrn>264375443725</rrn>
  <amount>2000.00</amount>
  <currencyCode>UGX</currencyCode>
  <tranType>WalletToAccount</tranType>
  <receiverMsisdn>256702837908</receiverMsisdn>
  <receiverName>NABAKOOZA DEBORAH</receiverName>
  <senderName>Test Customer</senderName>
  <senderMsisdn>256788566300</senderMsisdn>
  <callbackURL>http://10.11.6.59:7007/mfmbisug/mbintf/mtnCallbackService.jsp</callbackURL>

  <signature>QJmj6JGeznXt06YrYtpOFp+4jwn3mwUgal3ySLkoVf3PcAOehVyrC9v8qqjrj8L8x9i1zxW0v
ya+gpa+ltHVB58wjmTJdKKB9y+qDi1DFfaBlpqYthO1Oksu0QNkWwaGWy3hc5UgXS53Nh8yLNKsrE
C5KoCZ2wld6coOvoDIMLts5v2qEJ0HwZ7LuU4aKi2GwvV6Xbz+MXwLzv4f/gV92oBhDjsoa51kIN5kw
cz0nOahJLaIkchFD3gC7JLdrS8Yfhndy66JUfxmEBU5ZQnToUDxfZpSO8pwCxlTPoSWCGpVjpncE
acZMemSXkYVzXkqrLGzqSHoAV/NqdlhfnUVfg==</signature>
</EazzyDepositReq>
```

2.0.5 Sample Debit Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<EazzyDepositResp>
  <msgId>264375443725</msgId>
  <timestamp>2022-08-26T15:39:23.257</timestamp>
  <rrn>264375443725</rrn>
  <status>ACK</status>
  <responseDesc>Request Received By Telco. Waiting Final Status. Initial Telco Status:
    PENDING</responseDesc>
  <transactionRef>4580503</transactionRef>
</EazzyDepositResp>
```

3.0 TRANSACTION STATUS CHECK SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/status-check
Headers	Content-Type:application/xml

3.0.1 Transaction Status Check Request Fields And Description

NO	FIELD	DESCRIPTION	MANDATORY	TYPE
1	rrn	Retrieval reference number (Can be same as msgld)	Yes	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Yes	String
3	msgld	Reference number	Yes	String
4	telcom	Name of the destination Telco Example: MTN-UG	Yes	String
5	channel	Channel code Example: MFS	Yes	String
6	signature	Digital signature. See Digital signature section for more information.	Yes	String

3.0.2 Transaction Status Check Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	msgld	Reference number	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	String
3	rrn	Retrieval reference number	String
4	status	Transaction status	String
5	responseDesc	Transaction status description	String
7	transactionRef	Unique identifier for the transaction	String

3.0.3 Sample Transaction Status Check Request

```
<?xml version="1.0" encoding="UTF-8"?>
<transactionStatusInquireReq>
  <rrn>179003244933</rrn>
  <timestamp>26-Aug-2022 12:55:32</timestamp>
  <msgld>179003244933</msgld>
  <telcom>MTN-UG</telcom>
  <channel>MFS</channel>
  <signature
>QmxK1FFcHwCwylhplvCi7V4SLPTMO9mYaXxgLx4+dWTxetKLdkFSD6SpF5gaPER41+
</signature>
</transactionStatusInquireReq>
```

3.0.4 Sample Transaction Status Check Response

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<EazzyDepositResp>
  <msgld>179003244933</msgld>
  <timestamp>2022-08-29T11:55:50.085</timestamp>
  <rrn>179003244933</rrn>
  <status>PENDING</status>
  <responseDesc>Request Timeout Or Service Not Available At The Moment</responseDesc>
  <transactionRef>4580512</transactionRef>
</EazzyDepositResp>
```

3.0.5 Transaction Status Check Possible Values

STATUS(status)	DESCRIPTION
SUCCESS	Debit request was accepted by MTN awaiting the final status
FAILURE	Debit request failed
PENDING	Transaction final status is still not known
EXPIRED	Transaction is expired

4.0 TRANSACTION REFUND/ REVERSAL SERVICE

Method	POST
URL	https://10.11.6.111:8080/api/v1/refund
Headers	Content-Type:application/xml

4.0.1 Transaction Refund Request Fields And Description

NO	FIELD	DESCRIPTION	MANDATORY	TYPE
1	rrn	Retrieval reference number (Can be same as msgld)	Yes	String
2	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	Yes	String
3	telcom	Name of the destination Telco Example: MTN-UG	Yes	String
4	channel	Channel code Example: MFS	Yes	String
5	signature	Digital signature. See Digital signature section for more information.	Yes	String

4.0.2 Transaction Refund Response Fields And Description

No	FIELD	DESCRIPTION	TYPE
1	rrn	Retrieval reference number	String
2	msgld	Reference number	String
3	timestamp	Time of the transaction. Format: dd-mon-yyyy hh:mm:ss Example: 26-Aug-2022 12:55:32	String
4	amount	Original transaction amount	String
5	currencyCode	Transacting currency	
6	status	Transaction status	String
7	responseDesc	Transaction status description	String
8	transactionRef	Unique identifier for the transaction	String

4.0.3 Sample Transaction Refund Request

```
<?xml version="1.0" encoding="UTF-8"?>
<transactionRefundReq>
  <rrn>179003244933</rrn>
  <timestamp>26-Aug-2022 12:55:32</timestamp>
  <telcom>MTN-UG</telcom>
  <channel>MFS</channel>
  <signature
>QmxK1FFcHwCwylhplvCi7V4SLPTMO9mYaXxgLx4+dWTxetKLdkFSD6SpF5gaPER41+
</signature>
</transactionRefundReq>
```

4.0.4 Sample Transaction Refund Response

```
<?xml version="1.0" encoding="UTF-8"?>
<transactionRefundResp>
  <rrn>179003244933</rrn>
  <msgId>179003244933</msgId>
  <timestamp>26-Aug-2022 12:55:32</timestamp>
  <amount>2000.00</amount>
  <currencyCode>UGX</currencyCode>
  <status>SUCCESS</status>
  <responseDesc>Reversal Successful</responseDesc>
  <transactionRef>4580512</transactionRef>
</transactionRefundResp>
```

4.0.5 Transaction Refund Possible Values

STATUS	DESCRIPTION
SUCCESS	Reversal was successful from MTN
FAILURE	Reversal failed/ Transaction in failed final state
REVERSED	Transaction was previously reversed
PENDING	Transaction final status is still not known. Confirm final status first

5.0 DIGITAL SIGNATURE GUIDE

For the purpose of ensuring data integrity and security, every channel will be required to generate a digital certificate and share the same with the maintainer of this API. The certificate will be used to verify the signature contained in each payload.

5.0.1 *Generating The Signature Clear Text*

Each request should have a digital signature generated as described below:

1. **Generate the clear text:**

Concatenate the values in each request field shown below to generate the clear text. The clear text value will be hashed.

a. **Clear Text For Name Check**

```
clearText = mobileNumber  
          + msgId  
          + telcom  
          + channel
```

b. **Clear Text For Debit**

```
clearText = msgId  
          + rrn  
          + timestamp  
          + amount  
          + currencyCode  
          + tranType  
          + receiverMsisdn  
          + senderMsisdn
```

Note: The amount is a big-decimal value rounded off to a scale of 2, Ceiling

c. **Clear Text For Transaction Check Status**

```
clearText = rrn  
          + msgId  
          + telcom  
          + channel
```

d. **Clear Text For Transaction Refund**

```
clearText = rrn  
          + timestamp  
          + telcom  
          + channel
```

5.0.2 Sign The Clear Text

- Generate RSA private key
 - Using the private key and SHA-256 hashing algorithm, sign the clear text generated above.
- In summary, the signature scheme should be SHA-256 with RSA

5.0.3 Sample Java Code

The figure below shows an example of how the signature can be generated using Java although the same can be achieved using any programming language

Example of how to generate a digital signature using Java

```
import java.security.PrivateKey;
import java.security.Signature;
import java.util.Base64;

public class SignMsg {
    private static final String UTF_8 = "UTF-8";
    private static final String ALGORITHM = "SHA256WithRSA";

    public String signMessage(String clearText , PrivateKey privatekey) {
        try {
            byte[] messageBytes = clearText .getBytes(UTF_8);
            Signature signatureInstance = Signature.getInstance(ALGORITHM);
            signatureInstance.initSign(privatekey);
            signatureInstance.update(messageBytes);
            byte[] digitalSignature = signatureInstance.sign();
            String signature = Base64.getEncoder().encodeToString(digitalSignature);
            return signature;
        } catch (Exception ex) {
            ex.printStackTrace();
            return null;
        }
    }
}
```